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Lab 8 : “Potentiometer”

Introduction and objectives

This lab manual covers the basics of using the Analog-to-Digital Converter (ATD) module, focusing on a potentiometer connected to one of the analog input of MC9S12DT256 microcontroller.
The ATD module converts continuous analog voltage signals that vary between a min (0 V) and a max (5V) into discrete digital values represented in 8 or 10 bits resolution that can be stored, processed, and analyzed by the microcontroller.
During this lab session, you will learn how to configure and initialize the ATD module, perform analog-to-digital conversions and observe the digital output values.

Pre Lab 8 :

In this preliminary task, you are required to answer the following questions :

- MC9S12DT256 has 2 identical ATD (analog to digital converter) units , ATD0 and ATD1. Each ATD unit converts analog signals , connected to the analog inputs AN0→AN7 , to digital data , having 8 or 10 bits resolution , and received on ATD0DR0→ATD0DR7 16 bits data registers.
  A single 8 bits conversion can be performed as fast as 6 microsecond.
  The ATD0 associated registers are :
  4 <u>control registers</u> ATD0CTL2 → ATD0CTL5
  2 <u>status registers</u> ATD0STAT0 and ATD0STAT1
  8 <u>DATA registers</u> ATD0DR0 → ATD0DR7

By referring to the 'HCS12' datasheet located in 'Microcontroller' folder, make a brief explanation of the role of these registers , focusing on the underlined bits.

- ATD0CTL2 (ADPU, AFFC)
- ATD0CTL3 (S8C → S1C, FIFO)
- ATD0CTL4 (SRES8, SMP1, SMP0)
- ATD0CTL5 (DJM, DSGN, SCAN,MULT,CC,CB,CA)
- ATD0STAT0 (SCF)
- ATD0DR0

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Write fully commented programs for the PBMCUSLK board, including appropriate labels, memory operands and constants, to solve the tasks below:

Task 1:

The first program should do the following:

Turn ON LEDs ONE BY ONE when you turn the potentiometer all the way to the left , and then the Turn OFF the LEDs ONE BY ONE when you turn the potentiometer to the right .

Task2 :

The second program should do the following:

Check and convert continuously the value read from the potentiometer into a **voltage between 0.0 V and 5.0 V** *(with its tenths digit)* and display it on the LCD.

Task3:

The third program should do the following:

Check the converted digital value of the potentiometer every 2 s using **INTERNAL INTERRUPT** and turns ON the LEDs if this value exceeds 125.

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